

Munyan



American Welding Society
LEARNING

Homework

Book of Specifications Questions

(To be completed prior to attending AWS CWI Seminar)

April 3, 2016 Edition

CWI
Seminar

DISCLAIMER

The American Welding Society, Inc. assumes no responsibility for the information contained in this publication. An independent, substantiating investigation should be made prior to reliance on or use of such information.

International Standard Book Number:

American Welding Society

8669 NW 36 Street, #130, Miami, Florida 33166

© 2016 by American Welding Society

All rights reserved

Printed in the United States of America

Photocopy Rights. No portion of this document may be reproduced, stored in a retrieval system, or transmitted in any form, including mechanical, photocopying, recording, or otherwise, without the prior written permission of the copyright owner.

Authorization to photocopy items for internal, personal, or educational classroom use only or the internal, personal, or educational classroom use only of specific clients is granted by the American Welding Society provided that the appropriate fee is paid to the Copyright Clearance Center, 222 Rosewood Drive, Danvers, MA 01923, (978) 750-8400; www.copyright.com.

American Welding Society
Certified Welding Inspector Seminar

- **Visual Inspection Workshop** •

Homework:
Book of Specs
Practice Questions

April 3, 2016

Prepared by:
Ron Theiss, SCWI
Ed Norman, CWI

Table of Contents

Directions	7
Clause 1.0 General Requirements	9
Clause 2.0 Structural Steel	10
Clause 3.0 Pipeline	12
Clause 4.0 Pressure Piping	15
Clause 5.0 Procedure Qualification Requirements	17
Clause 6.0 Performance Qualification Requirements.....	18
Normative Annexes.....	21
Informative Annexes.....	23
Answer Key	25

Directions:

As of 2016, the Certified Welding Inspector Exam, in its Part B section, asks examinees to evaluate sample welds, welding situations and processes, welding symbols and WPSs and PQRs on the basis of a newly-formulated Book of Specifications. Although this Book of Specifications is similar in some places to several well-known welding codes, it is not identical with any, and has its own layout.

In order to efficiently respond to test questions on the the Part B portion of the CWI exam, it is imperative to have a working knowledge of the exam's Book of Specifications.

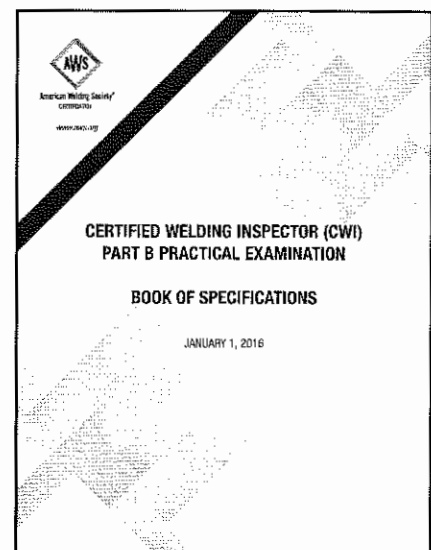
The questions in this homework assignment are intended to help you gain familiarity with the Part B Book of Specifications, so that when you attempt the exam, you'll be able to focus on the meaning of each question and not waste valuable time searching through the Book of Specifications for the needed paragraph, table or annex.

Please complete these practice items prior to the Thursday of your CWI Seminar week. On Thursday, you'll begin the Visual Inspection Workshop portion of the seminar. As judgments about the soundness of welds and the correctness of documentation and processes require reference to a standard, it will be assumed that you already have a good understanding of the requirements in the Book of Specifications.

All the questions in this homework assignment are answerable with reference to the Book of Specifications alone. No knowledge of any other code is required. The questions are grouped by the section of the Book of Specifications to which they pertain.

An Answer Key can be found at the end of this booklet.

**You will need a copy of the Certified
Welding Inspector (CWI) Part B Practical
Examination Book of Specifications
to complete this assignment**





Select any Question
in model Exam.

Use the CWI Part B Practical Examination Book of Specifications to answer the following Questions

Use the Book of Specifications Foreword and Clause 1.0 General Requirements to answer questions 1 – 5.

1. The Part B Book of Specifications includes specific clauses in three applications. These specific applications are:

- A. Aluminum, Stainless Steel and Carbon Steel
- B. Structural, Pressure Piping and Power Piping
- C. Structural Steel, Stainless Steel and Pressure Piping
- D. Structural, Pipeline and Pressure Piping
- E. Structural, Piping and Power Piping

2. In addition to the main body of the specification, there are:

- A. Annexes and tables
- B. Annexes, tables and a Book of Exhibits
- C. Annexes, tables and figures
- D. Annexes and figures
- E. Book of Exhibits and tables

3. The Annexes located in the Book of Specifications include:

- A. Normative – includes mandatory elements for use with this Specification
- B. Informative – is not part of this Specification but is included for informational purposes only
- C. Both A and B above
- D. Normative only
- E. Informative only

4. All fractional tolerances for fractional numbers from 1/16 inch but less than 1 inch are:

- A. $\pm 1/64$ inch
- B. $\pm 1/32$ inch
- C. $\pm 1/16$ inch
- D. $\pm 1/8$ inch
- E. $\pm 1/4$ inch

5. The fractional tolerances for whole numbers from 1 to 6 inches inclusive is:

- A. $\pm 1/64$ inch
- B. $\pm 1/32$ inch
- C. $\pm 1/16$ inch
- D. $\pm 1/8$ inch
- E. $\pm 1/128$ inch

Page 25 Answer Key

Use Clause 2.0 Structural, to answer questions 6 – 19.

6. When determining the overall length of a fillet weld:

- A. It is the length of the full size fillet weld including undersize starts
- B. It is the length of the full size fillet weld including undersize stops
- C. Both A and B above
- D. It is the length of the full size fillet weld excluding undersize starts
- E. It is the length of the full size fillet weld excluding undersize starts and stops

7. To determine the size of a fillet weld:

- A. It is the maximum size measured
- B. It is the maximum size measured plus the tolerance
- C. It is the minimum size measured
- D. It is the minimum size measured plus the tolerance
- E. It is the maximum size measured minus the tolerance

8. When terminating a fillet weld deposited on a cyclically loaded connection, the weld termination shall be made by:

- A. Holding the weld back from the corner a minimum of 1/2 inch
- B. Holding the weld back from the corner for a distance not less than the specified fillet weld size
- C. Wrapping the weld around the corner a minimum of 1 inch
- D. Wrapping the weld around the corner a minimum of the specified weld size and a maximum of two times the specified weld size
- E. Wrapping the weld around the corner for a distance not less than two times nor more than four times the specified fillet weld size

9. On all CJP groove welds in butt joints transverse to the computed tensile stress, they:

- A. Shall have no visible porosity that exceeds 1/32 inch
- B. Shall have no visible porosity that exceeds 3/32 inch
- C. Shall have no visible porosity
- D. Shall not have a sum of visible porosity that exceeds 3/8 inch
- E. Shall not have a sum of visible porosity that exceeds 1/2 inch

10. A 3/8 inch fillet weld was deposited on a lap joint on 1/2 inch steel plate. The width of the face of the fillet weld measures 9/16 inch. The maximum convexity permitted on this deposited fillet weld is:

- A. 1/16 inch
- B. 1/8 inch
- C. 3/16 inch
- D. None of the above – all fillet welds shall have a flat face
- E. None of the above – all fillet welds shall have a concave face

11. While performing a visual inspection you measure a mill-induced discontinuity to be 1 1/2 inches long and 1/8 inch deep. What action shall be taken concerning this mill-induced discontinuity?

- A. It shall be referred to the engineer for disposition
- B. It shall be referred to the welder for disposition
- C. It need not be repaired and the depth need not be explored
- D. It need not be repaired but the depth shall be explored
- E. It need not be repaired but the depth should be explored

12. With all fillet welds, overlap

- A. Shall be kept to a minimum
- B. Shall have a maximum length of 1/4 inch
- C. Shall have a maximum length of 1/8 inch
- D. Shall have a maximum length of twice the fillet weld size
- E. Shall be prohibited

13. The contract drawing indicates a 1/2 inch fillet weld 4 feet long. During your inspection of the fillet weld you measure 2 1/2 feet of the weld to be 7/16 inch. The fillet weld is:

- A. Unacceptable – 7/16 portion exceeds 10% of the weld length
- B. Unacceptable – entire weld shall measure 1/2 inch
- C. Undeterminable – type of connection was not given
- D. Acceptable – 1/2 inch given indicates the maximum weld size; there is no minimum
- E. Acceptable

14. When SAW is to be used, the edge preparation for weld surfaces shall not be rougher than:

- A. Sample 1 on the C4.1-77 Surface Roughness Guide
- B. Sample 2 on the C4.1-77 Surface Roughness Guide and shall have no gouges deeper than 1/8 inch
- C. Sample 3 on the C4.1-77 Surface Roughness Guide and shall have no gouges deeper than 1/8 inch
- D. Sample 4 on the C4.1-77 Surface Roughness Guide and shall have no gouges
- E. Sample 3 on the C4.1-77 Surface Roughness Guide and shall have no gouges

15. A structure is being fabricated out of 5/8 inch thick ASTM A 516 Grade 70 material. The welding process is FCAW and all welding will be performed in either the horizontal or vertical position. All surfaces that will receive welds shall have:

- A. No gouges
- B. No gouges deeper than 1/16 inch
- C. No gouges that exceed 1/32 in depth
- D. No gouges deeper than 1/8 inch
- E. No gouges that exceed 250 microinches

16. The maximum weld reinforcement for a CJP single V groove weld on 2 1/2 inch ASTM A 514 joined to 2 inch ASTM A 572 material in a cyclically loaded structure is:

- A. 1/16 inch
- B. 3/32 inch
- C. 1/8 inch
- D. 5/32 inch
- E. 3/16 inch

17. Arc strikes:

- A. Shall be ground to a smooth contour
- B. Shall be ground to a smooth contour and inspected using magnetic particle method
- C. Shall be ground to a smooth contour and inspected using the liquid penetrant method
- D. Shall be ground to a smooth contour and checked to ensure soundness
- E. Are prohibited

18. Acceptance criteria for ASTM A 709 Grade 100 steel shall be based on nondestructive testing performed not less than:

- A. 24 hours after the completion of the weld
- B. 48 hours after the completion of the weld
- C. 12 hours after the completion of the weld
- D. 8 hours after the completion of the weld
- E. Testing may begin immediately after the weld has cooled to ambient temperature

19. On all completed welds:

- A. Spatter shall be kept to a minimum
- B. Spatter is prohibited
- C. Spatter is allowed with SMAW only
- D. Spatter is acceptable for visual inspection only
- E. Spatter is acceptable on weld but unacceptable on parent metal

Use Clause 3.0 Pipeline, to answer questions 20 – 40.

20. For production welds, at no time shall the crown surface be raised above the parent metal by more than:

- A. 1/32 inch
- B. 1/16 inch
- C. 3/32 inch
- D. 1/8 inch
- E. No reinforcement is allowed

21. When inspecting a pipe weld on 8 inch schedule 40 (.322 wall) you mechanically measure some undercut at the face of the cover pass to be 0.015 inch deep and 3 inches long. The undercut is:

- A. Not acceptable
- B. Acceptable
- C. Not acceptable because length of undercut exceeds 2 inches
- D. Not enough information given
- E. Acceptable only if length of undercut is less than 6% of the length of the weld

22. CP that occurs in the finish pass shall be considered a defect if:

- A. The aggregate length of CP in any continuous 12 inch length of weld exceeds 2 inches
- B. Individual indications of CP each greater than 1/4 inch in length, are separated by less than 2 inch
- C. The size of any individual pore exceeds 1/16 inch
- D. The width of CP exceeds 1/16 inch
- E. The diameter of the cluster exceeds 1/2 inch

23. Cracks may be repaired provided a qualified repair welding procedure is used and the length of the crack is:

- A. More than 6% of the weld length
- B. More than 8% of the weld length
- C. More than 10% of the weld length
- D. More than 12.5% of the weld length
- E. None of the above



24. Inadequate penetration without high-low (IP) found by visual inspection shall be unacceptable when:

- A. The aggregate length of IP indications exceeds 3 inches in any continuous 12 inch length of weld
- B. The aggregate length of IP indications exceeds 2 inches in any continuous 12 inch length of weld
- C. The aggregate length of IP indications exceeds 1 inch in any continuous 12 inch length of weld
- D. The aggregate length of IP indications exceeds 1/2 inch in any continuous 12 inch length of weld
- E. The aggregate length of IP indications exceeds 1/4 inch in any continuous 12 inch length of weld

25. Cracks in welds may be repaired provided the following criteria are met:

- A. The length of the crack is less than 8% of the weld length
- B. The crater crack must be ground smooth
- C. A qualified repair welding procedure is used
- D. A and C above
- E. B and C above

26. An AI indication is:

- A. The Acceptance/Indication Standard
- B. The Authorization for Inspection
- C. Accumulation of Imperfections
- D. Acceptable Indication
- E. Angle of Incidence

27. Which of the following types of defects require company authorization?

- A. Crack repairs
- B. Double repairs
- C. Excessive undercut in the cover pass
- D. Porosity in the cover pass
- E. A and B above

28. For pipe ends of the same nominal wall thickness, the alignment offset shall not exceed:

- A. 1/4 inch
- B. 3/16 inch
- C. 1/32 inch
- D. 1/8 inch
- E. Thickness of pipe wall

29. A portion of the root bead where excessive penetration has caused the weld puddle to be blown into the pipe is known as:

- A. Inadequate penetration
- B. Internal concavity
- C. Excessive reinforcement
- D. Burn-through
- E. Hollow bead

30. The maximum depth for an unrepaired arc burn is:

- A. 1/64 inch
- B. 1/32 inch
- C. 1/16 inch
- D. 3/32 inch
- E. Arc burns are unacceptable

31. Excluding incomplete penetration due to high-low and undercutting, any AI shall be unacceptable if:

- A. The aggregate length of indications in any continuous 10 inch length of weld exceeds 2 inches
- B. The aggregate length of indications does not exceed 8% of the weld length
- C. The aggregate length of indications in any continuous 12 inch length of weld exceeds 10% of the weld length
- D. The aggregate length of indications exceeds 10% of the weld length
- E. None of the above

32. After excavation of a defect, the groove area shall be:

- A. Visually examined
- B. Examined by PT or MT
- C. Examined by RT
- D. A and B above
- E. A and C above

33. When one edge of the root bead (internal) is exposed because the adjacent pipe end is misaligned, this condition is known as:

- A. Internal undercut (IU)
- B. Incomplete fusion (IF)
- C. Inadequate penetration due to high-low (IPD)
- D. Incomplete penetration due to misalignment (IPDM)
- E. Incomplete filling at root (IFR)

34. When visually inspecting a weld on 6 inch schedule 40 pipe you measure an elongated single pore (porosity) to have a maximum length of 5/64 inch. The porosity is:

- A. Acceptable – does not exceed 1/8 inch
- B. Acceptable – does not exceed 1/4 inch
- C. Not acceptable – porosity cannot exceed 1/32 inch
- D. Not acceptable – porosity exceeds 25% of wall thickness
- E. Not acceptable – porosity exceeds 12.5% of wall thickness

35. Grinding repairs:

- A. Are limited to 2 inches in any continuous 12 inch length of weld
- B. Are limited to 8% of the weld length in any weld less than 12 inches in length
- C. Are required to have a qualified repair procedure
- D. Are to be made on internal defects only
- E. Do not require a repair procedure

36. All finished beads shall be:

- A. Started and stopped in the same location
- B. Flush with the parent metal
- C. Not started and stopped in the same location
- D. No wider than 5 times the size of the electrode
- E. Both A and B above

37. Rework:

- A. Does not require Company authorization
- B. Is considered a repair
- C. Requires a rework repair procedure
- D. Is prohibited
- E. Both B and C above

38. All completed welds and weld areas shall be:

- A. Free of all spatter
- B. No more than 1/8 inch wider than the width of the original groove
- C. Free of arc burns
- D. All of the above
- E. A and B above

39. Repairs to shallow crater cracks:

- A. Require a qualified repair procedure
- B. Are prohibited
- C. May be repaired by any abrasive method
- D. Shall be repaired by grinding only
- E. Require complete removal of the weld

40. A double repair of a crack:

- A. Is prohibited
- B. Shall have a repair procedure
- C. Shall have a double repair – reweld procedure
- D. Is allowed if the crack length is less than 8% of the length of the weld
- E. Is allowed if the crack depth is less than 1/2 of the wall thickness



Use Clause 4.0 Pressure Piping, to answer questions 41 – 51

41. Cracks in tack welds:

- A. Are acceptable if the entire tack weld is re-melted into the weld
- B. Shall not exceed 3/8 inch in length
- C. Are acceptable if in root of joint
- D. Are unacceptable
- E. Are acceptable with steel backing only

42. Overlap found in all as-welded surfaces:

- A. Is limited to 1/16 inch in length
- B. Is limited to vertical welds only
- C. Is limited to 3/8 inch in length
- D. Is unacceptable
- E. Is acceptable when using SMAW only

43. All discontinuities that result in a reduction of base metal thickness at the weld toe(s):

- A. Shall not exceed 8% of the pipe wall thickness
- B. Shall not exceed 1/64 inch in depth
- C. Shall not exceed 1/32 inch in length
- D. Shall not exceed 1/32 inch in depth
- E. Are prohibited

44. A weld repair was made on 8 inch schedule 80 pipe. The depth of the repair was measured to be 3/32 of an inch. The weld repair shall:

- A. Be inspected by visual inspection
- B. Be inspected by RT and PT or MT
- C. Be inspected by PT or MT
- D. Be inspected by PT or MT before postweld heat treatment
- E. Both A and B above

45. A single welded butt joint is applied to 6 inch schedule 80 to 6 inch schedule 120 pipe. The design temperature is 425°F. The maximum reinforcement allowed is:

- A. 1/16 inch
- B. 3/32 inch
- C. 1/8 inch
- D. 5/32 inch
- E. 3/16 inch

46. Rounded indications are indications that:

- A. Have a length more than three times their width
- B. Have a length less than three times their width
- C. Are always circular
- D. Are never circular
- E. Have a width 1/4 of their length

47. When internal misalignment measures 3/32 inch and trimming is used, the change in contour shall not exceed:

- A. 10 degrees
- B. 20 degrees
- C. 30 degrees
- D. 45 degrees
- E. Trimming is not preferred

48. Welding is being performed on 5 inch schedule 80 pipe. The minimum preheat temperature was applied properly before welding started and has been maintained during welding. 3/32 inch weld metal was deposited before the welding was interrupted and allowed to slowly cool. This condition is:

- A. Unacceptable because a minimum of 3/8 inch deposited weld metal is required
- B. Unacceptable because preheat temperature must be maintained until the joint is completed
- C. Acceptable because 3/32 inch is 25% of groove
- D. Unacceptable because 25% of groove is 5/32 inch
- E. Acceptable because 3/32 inch is more than 3/8 inch

49. A linear indication is found on the weld face and measures 5/32 inch in length. CAC-A was used to excavate the discontinuity. Immediately after excavation:

- A. VT is performed to assure complete removal of the discontinuity
- B. MT or PT is performed to assure complete removal of the discontinuity
- C. RT or UT is performed to assure complete removal of the discontinuity
- D. CAC-A is not permitted
- E. No excavation was necessary. Linear discontinuities shall not exceed 3/8 inch

50. After welding has been completed and the weld and weld area (including parent metal) have been thoroughly brushed and cleaned:

- A. All spatter on base metal surfaces shall be kept to a minimum
- B. All spatter on the base metal surfaces shall be separated by a minimum of 1/16 inch
- C. Spatter is only permitted when a semi-automatic welding process is used
- D. Base metal surfaces shall be free of spatter
- E. Spatter is not considered a discontinuity, therefore is not addressed in pressure piping

51. Two sections of 8 inch schedule have been fitted and welded. Before welding commenced, it was noted that there was internal misalignment that measured 1/16 inch. When measuring the thickness of the face reinforcement:

- A. The thickness shall be determined from the lower of the abutting surfaces
- B. The thickness shall be determined from the higher of the abutting surfaces
- C. The thickness shall be determined by the midpoint of the lower and higher surfaces
- D. Thickness measurements shall be taken from the upstream side of the weld
- E. Thickness measurements shall be taken from the downstream side of the weld

Use Clause 5.0 Procedure Qualification Requirements to answer questions 52 – 58

52. A welding procedure was qualified using SMAW for welding ASTM A 333 Grade 4 to ASTM A 333 Grade 4. This procedure would also qualify:

- A. Welding ASTM A 333 Grade 4 to ASTM A 333 Grade P12
- B. Welding ASTM A 333 Grade 4 to ASTM A 312 TP11
- C. Welding ASTM A 333 Grade 4 to API 5L Grade X52
- D. All of the above
- E. A and C above

Use the following information to answer questions 53, 54 & 55

A groove weld welding procedure was qualified using GMAW-S on 3/8 inch thick plate.

53. The maximum thickness of base metal qualified is:

- A. 0.750 inch
- B. 0.375 inch
- C. 0.4125 inch
- D. 0.625 inch
- E. 1 inch

55. Undercut shall not exceed:

- A. 0.0156 inch
- B. 0.0375 inch
- C. 0.0312 inch
- D. 8% of the base metal thickness
- E. Both B and C above

54. The bend tests required to qualify this procedure are:

- A. Two root bend specimens
- B. Two face bend specimens
- C. Four side bend specimens
- D. Both A and B above
- E. Either A & B or C above

56. A welding procedure is being qualified on 1 inch ASTM A333 Grade 9 material. The tensile specimen broke in the base metal outside the weld and weld area. The tensile strength was determined to be 59,850 PSI. The tensile specimen is:

- A. Acceptable
- B. Unacceptable
- C. Tensile tests are not required
- D. Not enough information provided
- E. No criteria provided

57. A welding procedure was qualified on pipe using an E7016 electrode. Because the addition of iron powder allows for welding of higher currents which result in higher deposition rates and deposition efficiencies the decision was made to weld with E7014 electrodes. Does this change of electrodes require a new WPS and a new PQR?

- A. No, both are SMAW electrodes
- B. No, both have the same AWS Specification
- C. No, both have the same F-Number
- D. Yes, each has a different F-Number
- E. Yes, each has a different AWS Specification

58. A welding procedure was qualified on 8 inch schedule 40 API 5L Grade X60 material. The bend specimens were thermally cut for root and face bends. The specimens were cut to a width of 1 1/2 inches with a length of 9 inches and the edges were rounded to a 1/8 inch radius using a grinder. The preparation of the bend specimens is:

- A. Acceptable – in compliance with Annex IV
- B. Unacceptable – width shall be 1 inch in width
- C. Unacceptable – length exceeds 9 inches
- D. Unacceptable – specimens shall be machined
- E. Unacceptable – cannot grind radius

Use the Book of Specifications Clause 6.0 Performance Qualification Requirements to answer questions 59 – 66

59. A welder qualifies on 1 inch ASTM A 516 Grade 55 plate using the FCAW-G process in the horizontal position. The maximum deposit thickness the welder is qualified for is:

- A. Twice the thickness of weld deposit including reinforcement
- B. Twice the thickness of weld deposit excluding reinforcement
- C. 1.1 times the thickness of weld metal deposited
- D. Unlimited
- E. 1 inch



Use the following information to answer questions 60, 61 & 62

A welder qualifies on 1/2 inch ASTM A 572, Grade 50 plate using the GMAW welding process, 0.035 inch dia. ER70S-4 electrode in the 4G position.

60. You mechanically measure an area of undercut to be 3/64 inch deep and 1 1/2 inches long at the toe of the face on the finished welded plate. The welded test plate is:

- A. Rejectable – undercut exceeds 1/32 inch
- B. Rejectable – undercut exceed 10% of the base metal thickness
- C. Rejectable – no undercut is acceptable
- D. Acceptable
- E. Both A and B above

61. If the test plate passed all the required tests, what is the most positions in which the welder is qualified to weld?

- A. Plate and pipe groove welds with an OD over 24 inches in the F and O positions and fillet welds in the F and O positions
- B. Plate and pipe groove welds with an ID over 24 inches in the F and O positions and fillet welds in the F and O positions
- C. Plate and pipe groove welds with an OD over 24 inches in the F, H and O positions and fillet welds in the F, H and O positions
- D. Plate and pipe groove welds with an OD over 24 inches in the F and O positions and fillet welds in the F, H and O positions
- E. All positions

62. The completed test groove weld plate:

- A. Shall be visually examined
- B. Shall be visually examined for discontinuities and should be aided by a strong light and magnifiers
- C. Shall receive both VT and RT before performing mechanical test
- D. May have UT substituted for VT
- E. May have RT substituted for VT

63. You visually inspected a completed test plate.

The only discontinuities found were four individual pores (porosity) located on the face of the weld. You measured each of the four pores. Each pore measured 3/32 inch.

- A. The test plate passes the visual inspection
- B. The test plate fails the visual inspection
- C. Not enough information provided
- D. No acceptance criteria for pores
- E. Pores cannot be detected using VT

64. A welder was qualified using the SMAW

process. The qualification was performed on 3/8 inch plate with a 1/8 inch E7018 electrode in the 3G position using DCEP. The power source to be used in the fabrication shop is AC only.

- A. The welder shall be requalified using a power source capable of producing AC
- B. The welder shall be requalified using an AC electrode
- C. Both A and B above
- D. A change in type of current does not require requalification
- E. Not enough information provided

65. The number of bend tests required for performance qualification when welding on pipe in the 3G position is:

- A. 2
- B. 4
- C. Not applicable
- D. No bend tests required
- E. 1

66. A welder qualified in the 2FR test position is also qualified to weld in which of the following positions?

- A. F, H on pipe $\leq$ 24 inches OD
- B. F, H, V on plate and pipe fillet
- C. F, H, O on plate and pipe fillet
- D. F, H, V on pipe over 24 inches OD
- E. F, H on plate and pipe fillet

67. During a performance qualification test, the inspector verifies that the placement and deposition of the root bead and the filler bead are acceptable. However, after carefully examining the completed weld, the inspector determines that the welder does not have the skills necessary to deposit acceptable weld beads. What action shall be taken?

- A. The inspector instructs the welder to grind the finished weld
- B. The welder fails due to visual examination
- C. The welder fails the test, but another test is scheduled
- D. Terminate the test
- E. None

Use the Normative Annexes to answer questions 68 – 77

68. A Numbers are as-welded chemical analysis of the deposited weld. The range of nickel for A-No 8 is:

- A. 7.50 – 15.00
- B. 15.0 – 37.00
- C. 0.80 – 4.00
- D. 1.25 – 2.80
- E. No nickel present in chemical composition of A-No 8

69. A welding procedure is being qualified on 1 inch ASTM A 516 Grade 65 material. The welding process is FCAW. The electrode classification is E71T-1 and the AWS Electrode Specification is A 5.20. What is the F Number for this classification?

- A. F1
- B. F2
- C. F4
- D. F5
- E. F6

70. The UNS Number for ASTM A 516 Grade 70 is:

- A. K20747
- B. K2516
- C. S30403
- D. K02700
- E. K11516

71. The product form for ASTM A 312 TP316 is

- A. Plate
- B. Seamless pipe
- C. Welded pipe
- D. Forged pipe
- E. Both B and C above

72. ASTM A 333 Grade 9:

- A. Is plate with a tensile strength of 65 ksi and a yield strength of 37 ksi
- B. Is pipe with a tensile strength of 63 ksi and a yield strength of 46 ksi
- C. Is seamless pipe with a tensile strength of 75,000 psi and a yield strength of 25,000 psi
- D. Is welded pipe with a tensile strength of 75,000 psi and a yield strength of 25,000 psi
- E. Is a bar with a tensile strength of 58 ksi and a yield strength of 36 ksi

73. The thickness limitations for ASTM A 240 S30815 is:

- A. ≤ 0.75 inch
- B. $> 0.75 \leq 1.5$ inches
- C. $> 2.5 \leq 4.0$ inches
- D. $< 3/16$ inch
- E. There are no thickness limitations

74. When preparing transverse bend specimens, the corners:

- A. Shall have a minimum radius of 1/8 inch
- B. Shall have a minimum radius of 1/16 inch
- C. Shall have a maximum radius of 1/16 inch
- D. Shall have a maximum radius of 1/8 inch
- E. All corners shall be 90° angles

75. When a welding procedure was qualified with steel backing:

- A. The backing shall be left in place on all root bends
- B. The backing shall be left in place on all face bends
- C. The backing shall be removed flush with the surface
- D. The backing may be removed
- E. Both A and B above

76. When preparing longitudinal bend specimens, the width of the specimen:

- A. Shall be a maximum of 1 inch
- B. Shall be 3/4 inch minimum
- C. Shall not be less than 3/8 inch
- D. Shall have a width of 1 1/2 inch
- E. No criteria given

77. If thermal cutting is used in the preparation of side bend specimens:

- A. Surface roughness shall not exceed sample 2 on the C4.1-77 guide
- B. Surface roughness shall not exceed sample 3 on the C4.1-77 guide
- C. Allow 1/8 inch to be machined from each thermal cut surface
- D. Machine 1/8 inch from each thermal cut surface in weld area only
- E. Machine 1/8 inch from side that will be in tension

Use the Informative Annexes to answer questions 78 – 85

78. The abbreviation CVN stands for:

- A. Controlled Volumetric Nominal
- B. Cluster Visual Numbering
- C. Complete Verification of Nondestructive
- D. Charpy V-notch
- E. Conditions Variable of Nondestructive

79. The abbreviation PJP stands for:

- A. Proper Joint Preparation
- B. Preliminary Joules Projected
- C. Preliminary Joint Procedure
- D. Prequalified Joint Preparation
- E. Partial Joint Penetration

80. The abbreviation TYP stands for:

- A. Typing error
- B. Tensile & Yield Properties
- C. Temperature Yippers Procedure
- D. Typical
- E. Toughness & Yield Properties

81. The decimal equivalent of 6/64 is:

- A. 0.046
- B. 0.078
- C. 0.093
- D. 0.156
- E. 0.171

82. 4 inch schedule 40 has a nominal wall thickness of:

- A. 0.203 inch
- B. 0.216 inch
- C. 0.226 inch
- D. 0.237 inch
- E. 0.258 inch

83. A machined feature on out-of-round pipe inside diameters to make sure inside diameters are in proper alignment for welding is referred to as:

- A. An inside bore
- B. An alignment bore
- C. A counter machining
- D. A proper bore
- E. A counterbore

84. For pipeline applications, during welding or after the weld has been completed, the removal of an imperfection that requires grinding and/or welding that is performed prior to VT or NDT of a completed weld is referred to as:

- A. Repair
- B. Reweld
- C. Rework
- D. Secondary repair
- E. Counterweld

85. A repair of a repair is referred to as:

- A. Rework
- B. Double rework
- C. Re-rework
- D. Double re-rework
- E. Double repair

Answer Key

Use the Book of Specifications Foreword and Clause 1.0 General Requirements to answer questions 1 – 5.

1. D – Foreword, Page i
2. C – Foreword, Page i
3. C – Table of Contents, Page ii
4. B – 1.3.2, Page 2
5. C – 1.3.2, Page 2

Use Clause 2.0 Structural Steel to answer questions 6 – 19.

6. E – 2.4.1, Page 4
7. C – 2.4.2, Page 4
8. E – 2.2.4.2, Page 4
9. C – Table 1, #8(A), Page 4
10. B Figure B, Page 7 – Tables 2 and 3, Page 6 –
11. E – 2.1.1, Page 3
12. E – 2.3.1.4, Page 4
13. E – Table 1, #6, Page 5
14. E – 2.2.1.3, Page 3
15. B – 2.2.1.1, Page 3
16. E – Figure A, Page 7, Table 2, Page 6 – Table 3, Page 6 (t = thickness of thicker plate joined)
17. E – 2.2.2, Page 3
18. B – Table 1 (5), Page 5
19. D – 2.2.3, Page 3

Use Clause 3.0 Pipeline to answer questions 20 – 40.

20. B – 3.1.4, Page 9
21. B – Table 4, Page 12 #3 depth – 0.015 inch or 6% of 0.322 inch = 0.019 inch
22. E – 3.2.5.3.1, Page 12
23. E – 3.1.9.2 (1), Page 9
24. C – 3.2.1.2, Page 10
25. D – 3.1.9.2, Page 9
26. C – 3.2.8, Page 12
27. E – 3.1.9.1, Page 9
28. D – 3.1.3, Page 9
29. D – 3.2.4, Page 12
30. E – 3.1.8, Page 9
31. E – 3.2.8, Page 12
32. D – 3.1.9.7, Page 10
33. C – 3.2.2, Page 11

34. D – 3.2.5.2, Page 12, Annex V, Page 45 – 25% of 0.280 inch = $0.07 - 5/64 = 0.078 - (0.078 > 0.07)$
35. E – 3.1.9.4, Page 10
36. C – 3.1.5, Page 9
37. A – 3.1.9.1, Page 9
38. D – 3.1.6, 3.1.7, 3.1.8, Page 9
39. C – 3.1.9.2, Page 9
40. A – 3.1.9.2, Page 9

Use Clause 4.0 Pressure Piping to answer questions 41 – 51

41. D – 4.1.4, Page 13
42. D – 4.1.7, Page 13
43. D – 4.2.2, Page 14
44. E – 4.1.9.2, Page 13 & 4.1.9.3, Page 14
45. D – 4.2.3, Page 14 NOTE 3, Annex VI, Page 47 (sch. 80 = 0.432 [7/16"] sch. 120 = 0.562 [9/16"] – Table 5, Page 14 = 5/32" (Note: Fractional equivalence can be found in Table 18, Page 45)
46. B – 4.2.7, Page 14
47. C – 4.1.1, Page 13
48. C – 4.1.5, Page 13 & Annex VI, Page 47
49. B – 4.1.9, Page 13
50. D – 4.1.8, Page 13
51. B – Table 5, Page 14 – Note 4

Use Clause 5.0 Procedure Qualification to answer questions 52 – 58

52. E – 5.2.1.1, Page 17, Annexes III-A and III-B (ASTM A 3333 Grade 4 is M #4) – (ASTM A 335 Grade P12 is M #4) – (API 5L Grade X52 is M #1)
53. C – Table 7 (b), Page 18
54. C – 5.3.1, Page 22 – Bold Paragraph, Page 22
55. C – 5.4.1.2, Page 22
56. A – 5.4.3.3, Page 23 & Annex III-B, Page 39, Minimum specified – 63,000. $5\% \text{ of } 63,000 = 3,150$; $63,000 - 3,150 = 59,850$
57. D – 5.2.3.3, Page 19 & Annex II, Page 33
58. A – 5.4.2, Page 22 – Annex III-B & Annex IV - (b), Page 41

Use Clause 6.0 Performance Qualification to answer questions 59 – 67

- 59. D – Table 10, Page 26
- 60. A – 6.4.1.2(3), Page 30
- 61. D – Table 15, Page 29
- 62. A – 6.4.1.1, Page 29
- 63. A – 6.4.1.2(5), Page 25
- 64. D – 6.2, Page 25
- 65. C – Table 12, Page 27
- 66. E – Table 15, Page 29
- 67. D – 6.4.1.1, Page 29

Use the Normative Annexes to answer questions 68 – 77

- 68. A – Annex I, Page 31
- 69. E – Annex II, Page 33
- 70. D – Annex III-A, Page 36
- 71. E – Annex III-A, Page 35
- 72. B – Annex III-B, Page 39
- 73. D – Annex III-B, Page 39
- 74. D – Annex IV, Page 41
- 75. C – Annex IV (c), Page 41
- 76. E – Annex IV, Page 41 & 42
- 77. C – Annex IV, Page 42

Use the Informative Annexes to answer questions 78 – 85

- 78. D – Annex V, Page 43
- 79. E – Annex V, Page 43
- 80. D – Annex V, Page 43
- 81. C – Annex V, Table 18, Page 45
- 82. D – Annex VI, Page 47
- 83. E – Annex X, Page 55
- 84. C – Annex X, Page 55
- 85. E – Annex X, Page 55